



Github and COPR

A CI/CD pipeline for Fedora Remix and utilities

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Overview

What are we doing?

- ▶ Git Repositories
- ▶ Build containers
- ▶ Github Actions and Workflows (pipelines)
- ▶ ~~Jenkins (pipelines)~~
- ▶ COPR repositories
- ▶ Github Pages
- ▶ Github Marketplace



Why are we doing this?

Continuous Integration(CI) & Continuous Delivery (CD)

A key DevOps principle for automation, consistency and reliability

Continuous Integration

Continuous Integration is an automation process targeting developers to improve the quality of their code and the reliability of their applications. Each Git commit is verified by an **automated build**, allowing teams to detect problems early.

Continuous Delivery

The goal of Continuous Delivery is to have code in a state that is always **ready for deployment** to a production environment.

Continuous Deployment

The goal of Continuous Deployment is to push all of your changes to your end users **as quickly and efficiently as possible**.

Slide Content Credit:
“Native Kubernetes CI/CD Pipelines”
by Álvaro López (Senior Consultant)
Antonio Arjona (Senior Consultant).



Problems and Solutions

Managing Large Files and Scripts

Pain Points and Integration Issues

- Embedded applications not tied to Fedora Remix existed in the repos
- Installation of embedded or custom applications overly complex
- Long kickstart and debugging process
- Testing of updated contents and distribution of just the updated application or scripts
- Requirements for build system and maintaining a system to build the Fedora Remix

Solutions and Improvements

- Modular applications
- Separate repositories (**GIT**)
- Installation repositories (**COPR**)
- Packaging of software/scripts as RPMs



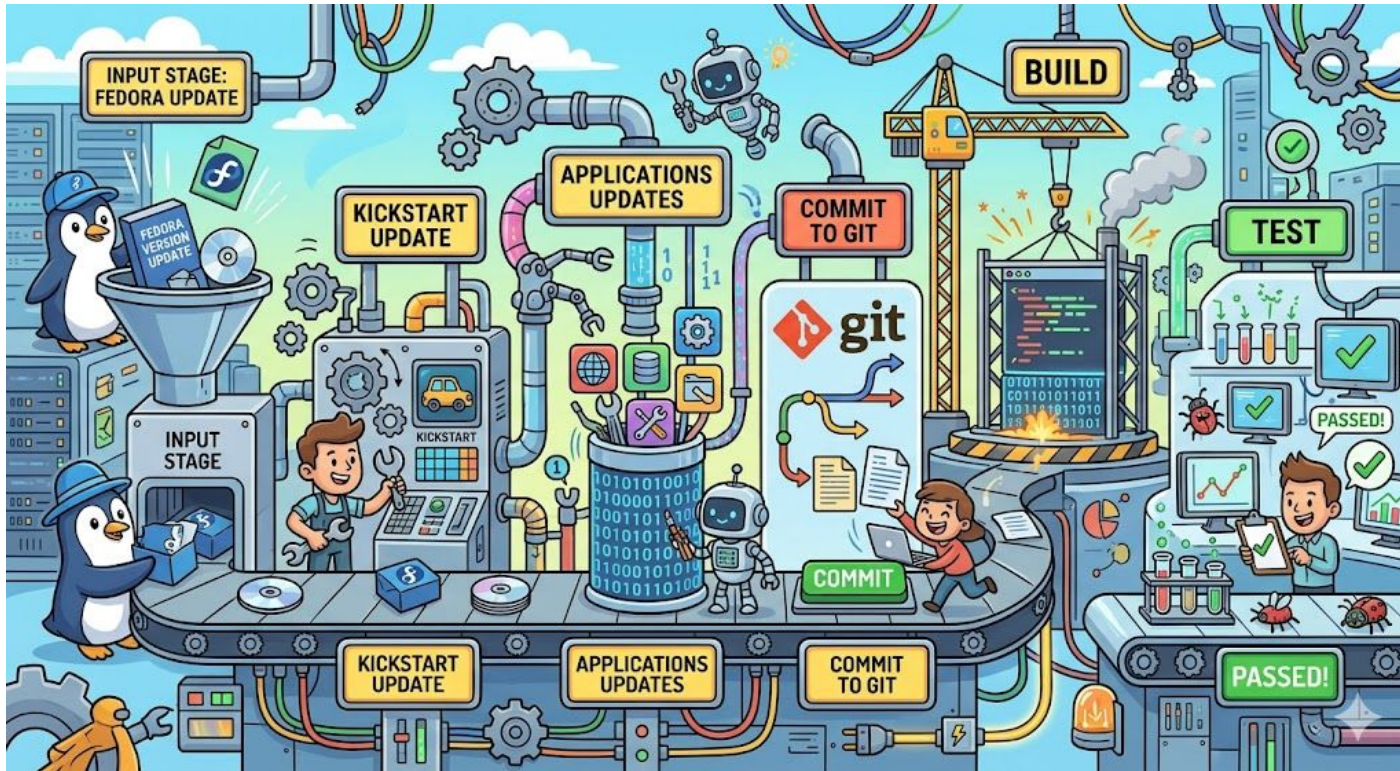
Repositories for Project Components

- https://github.com/tmichett/Fedora_Remix
- <https://github.com/tmichett/RemixBuilder>
- https://github.com/tmichett/Fedora_Remix_Lab
- https://github.com/tmichett/Fedora_Remix_Tools
- <https://github.com/tmichett/FedoraRemixPXE>
- <https://github.com/tmichett/FedoraRemixCustomize/>
- <https://github.com/tmichett/dyff>
- https://github.com/tmichett/log_viewer_copr
- https://github.com/tmichett/log_viewer

Repositories for RPM Packages

- <https://copr.fedorainfracloud.org/coprs/tmichett/FedoraRemix/packages/>

Providing Updates Quickly & Reliably



Issues

- Maintaining build system to match version of Fedora being built
- Maintaining versions on packages or custom scripts being supplies
- Long build process from VMs or physical machine after ensuring it met build requirements
- Documentation and PDFs

Solutions and Improvements

- Automated build process of applications
- Containerization and scripting of Fedora Remix ISO
- Automated RPM build process
- RPM installation of custom applications and scripts
- Github Marketplace Action

Automating and Creating Consistent Builds

Containers & Scripts

Simple Requirements

- Linux system with Podman installed and configured
- Github Personal Access Token for pushing to Github Container Registry (only required if you are building container image)
- SSH key for Github access and operations (used by container)
- Cloned copy of Fedora Remix Project in a directory on your local system
- Updated **config.yml** file for settings on the build
- **Update_Remix_Config.sh** - script to update the config.yml files
- **Verify_Build_Remix.sh** - script to verify config files and container is available

config.yml

```
# Interactive mode - shows menu of available kickstarts
./Build_Remix.sh

# Direct kickstart selection (skip menu)
./Build_Remix.sh -k FedoraRemixCosmic

# List available kickstart files
./Build_Remix.sh -l

# Show help
./Build_Remix.sh -h
```

Solution

Container_Properties:

```
Fedora_Version: "43" # Update this to match your target Fedora version
SSH_Key_Location: "~/ssh/github_id" # Path to your GitHub SSH key
Fedora_Remix_Location: "/path/to/your/output/directory" # Output directory for ISO build
GitHub_Registry_Owner: "your-github-username" # Your GitHub username or organization
# Image_Name is auto-generated from GitHub_Registry_Owner and Fedora_Version
# Format: ghcr.io/{GitHub_Registry_Owner}/fedora-remix-builder:{Fedora_Version}
```

How does it work?

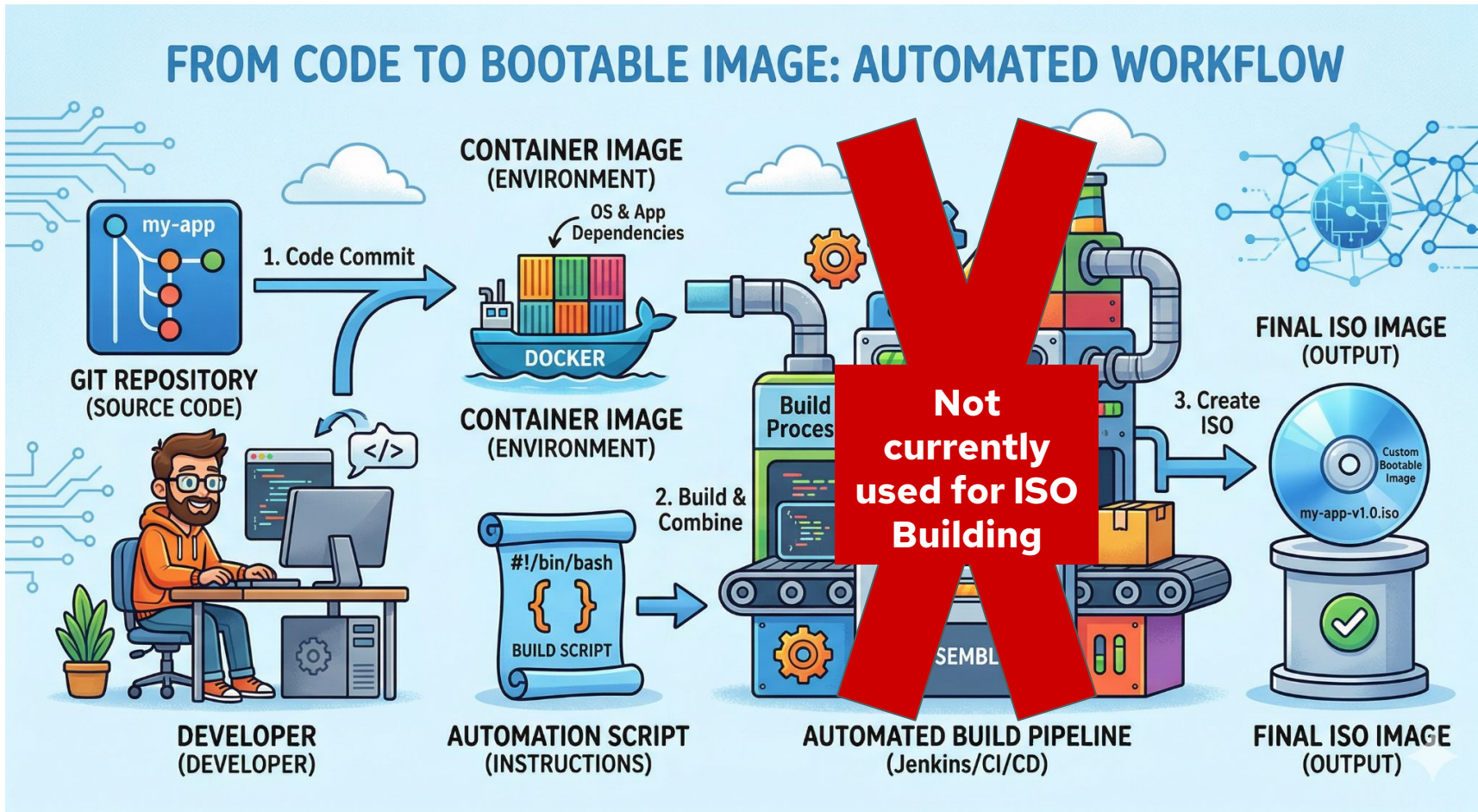
Core Components

What's Needed

- Fedora Remix Repo
- Fedora Remix Builder Image
- Automated RPM build process
- **Build_Remix.sh** build script

What Happens

- Clone Fedora Remix repository
- Make updates to kickstart files (mainly packages to be installed)
- Update various configuration files with versions
- **Build_Remix.sh** and **Verify_Build_Remix.sh** use Podman to build the ISO based on the configuration in the **config.yml** file
 - Menu available to select which Remix Kickstart being used to build



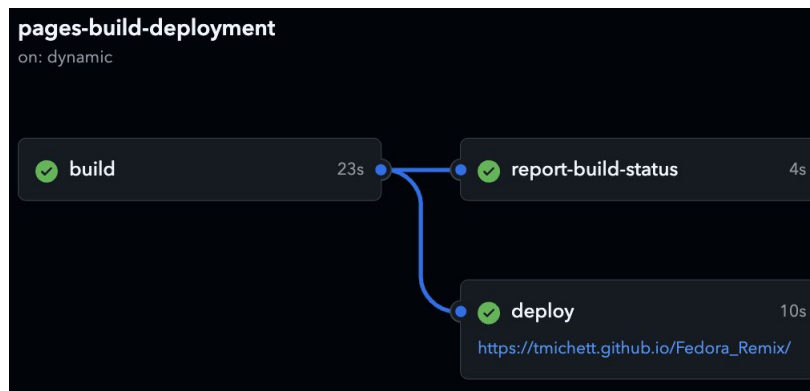
Modular Apps & Components

Kickstart Snippets (35+)

```
Fedora_Remix / Setup / Kickstarts / KickstartSnippets / format-functions.ks
tmichett FEAT: Refactor - Directory Rename from Ansible to ... 68c2410
11 lines (93 loc) · 4.31 KB
Code Blame Raw
1  ## Shared Formatting Functions for Kickstart Snippets
2  ## This file provides consistent formatting across all installation
3
4  # Set proper PATH to ensure all tools are available and prevent warn
5  export PATH=/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/
6
7  # Enable UTF-8 for Unicode support
8  export LANG=en_US.UTF-8
9  export LC_ALL=en_US.UTF-8
10
11 # Enhanced color definitions
12 readonly RED='\033[0;31m'
13 readonly GREEN='\033[0;32m'
14 readonly YELLOW='\033[1;33m'
15 readonly BLUE='\033[0;34m'
```

```
Fedora_Remix / Setup / Kickstarts / KickstartSnippets / setup-tmux.ks
tmichett FEAT: Refactor - Directory Rename from Ansible to ... 68c2410
5 lines (5 loc) · 123 Bytes
Code Blame Raw
1  ## Create TMUX Config Directory
2  echo "Configuring TMUX"
3  mkdir /opt/tmux
4  cd /opt/tmux
5  wget http://localhost/files/tmux.conf
```

Github Actions & Workflows (10+)



tmichett / log_viewer

<> Code Issues Pull requests

Actions

New workflow

All workflows

- Automated Comprehensive Release
- Build Flatpak Package
- Build macOS App - Dual Architecture
- Build macOS App - Intel
- Build RPM Package

Repositories and Registries (10+)

tmichett

Overview Repositories 434 Projects Packages 1 Stars 63

fedora-remix-builder 41

Public Latest

Install from the command line [Learn more about packages](#)

```
$ docker pull ghcr.io/tmichett/fedora-remix-builder:41
```

fedora copr

OIDC login | gssapi-login | sign up

Search projects by name, os or arch

Home > tmichett > FedoraRemix > Packages

tmichett / FedoraRemix

Project ID: 201137

Overview Packages Builds Monitor

Project Packages

Showing 1 to 3 of 3 items

Name	Last Build Version	Last Build Submitted	Last Build Status	Automatic Build	Actions
dyff	1.9.3-1	5 months ago	succeeded	Disabled	-
fedora_remix_tools	1.0-9	3 months ago	succeeded	Disabled	-
LogViewer	4.0.5-0	5 months ago	succeeded	Disabled	-

Show 25 entries

Possible build states

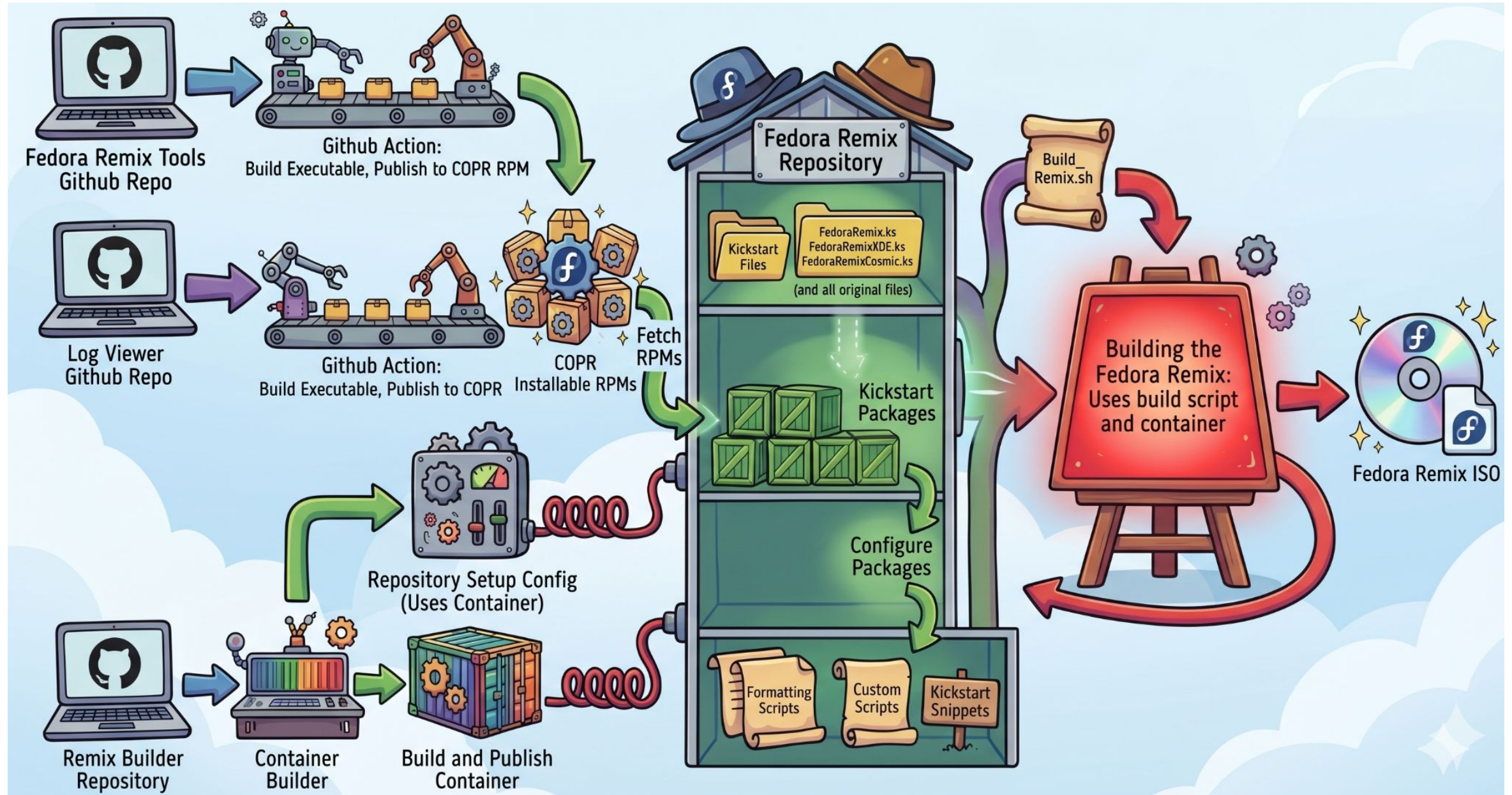
Marketplace / Actions

gh-md2pdf-action

release v1 marketplace gh-md2pdf-action build passing

A GitHub Action to convert Markdown files to PDF with support for Mermaid diagrams, GitHub-style markdown rendering, and automatic PDF bookmarks.

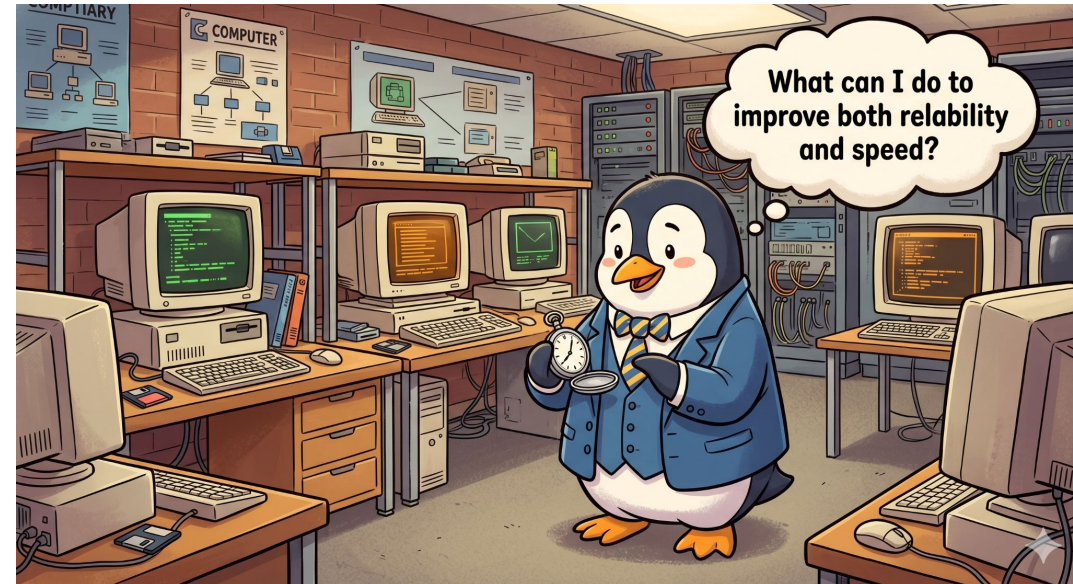
Putting it all together!



What Did We Learn

Recap *(Easy, Automatic, Repeatable, Reliable, Fast)*

- Smaller Repos and Projects
 - Automation releases continual and incremental updates
 - Easier management of component pieces and backward compatibility
 - Separate RPMs allows upgrading existing versions that are installed
- Containerized Build Process
 - Build on any system with minimal requirements
 - Quicker deployments of new versions
- Github Actions, Marketplace, and Pages
 - Automatic documentation deployment
 - Repeatable styles across projects and repositories
 - Faster time to deployment



Results of Builds

- Fedora 42 => 43 Release process (3-6 days)
 - Semi-Automated (Ansible/Python)
 - Github pages
 - Non-containerized (VM or Physical build only)
- Fedora 43 => 44 Release process (less than 2 hours)
 - Fully automated (Python/BASH)
 - Containerized
 - Github pages and Marketplace

More Information

Fedora Remix GitOps

Home

Quickstart: Container Build

Quickstart: Physical/VM Build

Fedora Remix

Remix Builder

Fedora Remix Customize

Fedora Remix Tools

Fedora Remix Lab

Log Viewer

gh-md2pdf-action

Search Fedora Remix GitOps

Dark

GitHub

Fedora Remix – GitOps & Automation

GitHub and COPR – Red Hat Summit 2026 Presentation Materials

Quickstart Guides

Get started building your own custom Fedora Remix ISO. The **Container Build** is the **recommended method** — it provides a consistent, reproducible environment across different systems without modifying your host.

Guide	Description
Container Build	Recommended. Build a Fedora Remix ISO using the containerized build system with Podman
Physical/VM Build	Build a Fedora Remix ISO directly on a physical or virtual machine





Thank you



[linkedin.com/company/red-hat](https://www.linkedin.com/company/red-hat)



[youtube.com/user/RedHatVideos](https://www.youtube.com/user/RedHatVideos)



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